

Executive Business Report

Executive Summary

The company generated \$2,261,537 in sales across 9,800 orders from 2015 to 2018. Sales increased over time, with Technology as the top category and the West and East regions as the strongest territories. The forecast points to continued sales in the next three months, while the anomaly check shows a small number of unusual weeks that should be reviewed before they affect planning. The main business action is to keep more stock ready for fast-moving items, protect the strongest regions and categories, and use the forecast as a planning guide rather than a fixed promise.

Key Findings from the Data

- Total sales reached \$2,261,537 across 9,800 orders.
- Technology led with \$827,456, followed by Furniture at \$728,659 and Office Supplies at \$705,422.
- The West region produced \$710,220 and the East region produced \$669,519, making these the two most important regions by revenue.
- Annual sales rose from \$479,856 in 2015 to \$722,052 in 2018, showing a clear upward trend.

3-Month Sales Forecast

The forecast model produced a validation error of about \$18,939 on average and about \$21,382 in typical monthly error. The table below translates that into a simple planning range for the next three months.

Month	Forecast sales	Planning range	Business meaning
January 2019	\$42,656	\$21,274 to \$64,038	Expected monthly sales
February 2019	\$24,604	\$3,223 to \$45,986	Expected monthly sales
March 2019	\$48,618	\$27,236 to \$69,999	Expected monthly sales

In plain language, the next three months are expected to be about \$42,656, \$24,604, and \$48,618. The likely planning band around those numbers is wide enough that purchasing should stay flexible.

Top 3 Anomalies Detected

The table below shows the three largest unusual weekly sales points and the most likely reason they happened.

Week	Sales	Likely cause
2015-03-22	\$37,704	Likely a large one-off order or promotion-driven spike.
2015-09-13	\$29,959	Likely a large one-off order or promotion-driven spike.
2015-07-26	\$21,590	Likely a large one-off order or promotion-driven spike.

Product Demand Segmentation

The products separate into four useful stocking groups. One group covers the biggest revenue drivers, one contains expensive but more variable items, one covers broad everyday demand, and one is a small niche group. The inventory policy should differ by group rather than using one blanket rule.

Segment	Sub-categories	Stocking strategy
High-volume core items (Cluster 2)	Accessories, Binders, Chairs, Phones, Storage, Tables	Keep strong safety stock and replenish often because demand is large and consistent.
Higher-value but more variable items (Cluster 0)	Copiers, Machines	Stock carefully and watch for spikes because these items are expensive

		and less predictable.
Broad everyday demand items (Cluster 1)	Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper	Use moderate stock and frequent checks because this group is steady but spread across many sub-categories.
Niche lower-volume items (Cluster 3)	Supplies	Keep lean inventory and order in smaller batches to avoid cash being tied up in slow movers.

Business Recommendations

- Put more inventory focus on Technology and the West/East regions because they delivered the strongest sales, with Technology at \$827,456 and the West region at \$710,220.
- Treat the forecast as a planning target and reserve flexibility for monthly swings, because the model still shows an average error of about \$18,939 and a typical error of about \$21,382.
- Review unusual weekly sales spikes before ordering, because the anomaly model flagged several weeks where sales moved far above the recent average; these could be legitimate large orders, but they can also distort inventory plans if left unchecked.

Risk / Limitation

The forecast and anomaly checks rely only on past sales data. They do not include promotions, stockouts, price changes, competitor actions, or other events that can change demand quickly. That means the system is useful for planning, but it should not be the only input for purchasing or budget decisions.